Positive rates do not force stationary measures to be quasilocal (on a tree)

counterexamples-ips, entry 0004

Status: *plausible* (finite identities machine-checked; not yet checked by a human)
Counterexample: ChatGPT Astra-6, 7 September 2026 (raw output in `raw/`)
Digest: Claude (Opus 5.5), 2 November 2026

Question. Consider an interacting particle system with finite local state space and finite-range transition rates that are uniformly bounded away from zero. Is every stationary measure μ quasilocal? The question is inspired by elliptic regularity: stationary measures solve $\mathscr{L}^*\mu = 0$, positive rates play the role of uniform ellipticity, and quasilocality (continuity of conditional probabilities in the product topology) the role of regularity. On $\mathbb{Z}^d$ the question is open. On the 3-regular tree the answer is *no*.

Setting. In the *directed binary tree* every vertex x has one parent $p(x)$ and two children; the undirected graph is the 3-regular tree, oriented towards a fixed end. Local state space $E = \{0, 1, 2\}$. On each oriented edge $(x, p(x))$ the ordered pair $(\eta_x, \eta_{p(x)})$ jumps according to one fixed 9-state generator $Q = (q_{ij})$ with the rates of Table 1, independently over edges. All 72 rates are positive, the smallest is $\varepsilon_* = 1265009/6511300000 \approx 1.9 \cdot 10^{-4}$, and all row sums are below 33.

	00	01	02	10	11	12	20	21	22
00	–	$\frac{2767709}{9766950000}$	$\frac{10385390113}{33207630000000}$	$\frac{589477}{1953390000}$	$\frac{62846}{203478125}$	$\frac{1813959}{6511300000}$	$\frac{4589347}{19533900000}$	$\frac{2536387}{9766950000}$	$\frac{204841}{1028100000}$
01	$\frac{9391756087}{33207630000000}$	–	$\frac{9595487813}{33207630000000}$	$\frac{4516031}{19533900000}$	$\frac{1545599}{6511300000}$	$\frac{1265009}{6511300000}$	$\frac{5577797}{19533900000}$	$\frac{5102083}{19533900000}$	$\frac{305591}{1028100000}$
02	$\frac{21707821}{74290000000}$	$\frac{13}{50000}$	–	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$
10	$\frac{13}{50000}$	$\frac{630401}{25000}$	$\frac{13}{50000}$	–	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{17029}{20000}$	$\frac{13}{50000}$
11	$\frac{13}{50000}$	$\frac{17809}{20000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	–	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$
12	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	–	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$
20	$\frac{216521}{50000}$	$\frac{13}{50000}$	$\frac{1411131}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	–	$\frac{13}{50000}$	$\frac{13}{50000}$
21	$\frac{34733}{100000}$	$\frac{13}{50000}$	$\frac{89301}{100000}$	$\frac{433037}{100000}$	$\frac{13}{50000}$	$\frac{5113}{2000}$	$\frac{13}{50000}$	–	$\frac{13}{50000}$
22	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{17039}{100000}$	$\frac{13}{50000}$	$\frac{13}{50000}$	$\frac{733}{100000}$	$\frac{104209}{25000}$	$\frac{13}{50000}$	–

Table 1: Off-diagonal rates q_{ij}; rows: from, columns: to; pair states ordered $00, 01, \ldots, 22$.

Theorem 1 (ChatGPT Astra-6)**.** *Let* $\pi^A = (\frac{23}{100}, \frac{14}{100}, \frac{63}{100})$ *and* $\pi^B = (\frac{19}{100}, \frac{31}{100}, \frac{1}{2})$*. The product measures* $\nu^A = \bigotimes \pi^A$ *and* $\nu^B = \bigotimes \pi^B$ *are both stationary. Hence* $\mu = \frac{1}{2}\nu^A + \frac{1}{2}\nu^B$ *is stationary, and* $\mu(\eta_o = 0 \mid \mathcal{F}_{\mathbb{V}\setminus\{o\}})$ *has no continuous version:* μ *is not quasilocal.*

Lemma 2 (adjoint identity)**.** *For a full-support product measure* ν_π *and the generator part* K_e *of an edge* $e = (x, p(x))$*,* $\int K_e f \, d\nu_\pi = \int f \, h_\pi(\eta_x, \eta_{p(x)}) \, d\nu_\pi$*, where*

$$h_\pi(a,b) = \frac{1}{\pi_a \pi_b} \sum_{(c,d)\neq(a,b)} \pi_c \pi_d \, q_{(c,d),(a,b)} - \sum_{(c,d)\neq(a,b)} q_{(a,b),(c,d)}.$$

Lemma 3 (tree-divergence criterion)**.** *If* $\sum_a \pi_a v(a) = 0$ *and* $h_\pi(a,b) = v(b) - 2v(a)$ *for all* a, b*, then* ν_π *is stationary.*

Proof. For f local with support in a finite A, only edges touching A contribute. Integrating out an endpoint outside A kills its term (mean zero). Each $y \in A$ is the child of one edge (term $-2v(\eta_y)$) and the parent of two edges ($+v(\eta_y)$ each), so $\int \mathscr{L} f \, d\nu_\pi = 0$. Local functions form a core (Liggett 1985, Ch. I). □

Finite identities. With $v^A = (-\frac{45787}{3680}, \frac{87917}{100000}, \frac{434699}{100000})$ and $v^B = (-\frac{12245721}{950000}, \frac{44491}{50000}, \frac{21733}{5000})$ the two mean-zero conditions and all 18 balance identities hold exactly (checked in exact rational arithmetic by `verify.py`; an independent brute-force check of $\int \mathscr{L} f \, d\nu = 0$ is `check_stationarity.py`; both in the entry's `code` folder).

Not quasilocal. The density of 0s outside o is $\frac{23}{100}$ under ν^A and $\frac{19}{100}$ under ν^B, so a tail event separates the components and $\mu(\eta_o = 0 \mid \mathcal{F}_{\mathbb{V}\setminus\{o\}})$ equals $\frac{23}{100}$ ν^A-a.s. and $\frac{19}{100}$ ν^B-a.s. Both sets are dense (full support), so a continuous version would have to be constant equal to both values.

Why a tree?

Proposition 4 (added in this digest)**.** *For a translation-invariant nearest-neighbour pair-update IPS on* $\mathbb{Z}$ *with irreducible edge generator* Q *on* E^2*, at most one full-support product measure is stationary.*

Proof. (i) On $\mathbb{Z}$ every vertex has one parent and one child; testing Lemma 2 on intervals shows that ν_π is stationary iff $h_\pi(a,b) = v(b) - v(a)$ for some v. (ii) Let $\nu_\pi, \nu_{\pi'}$ be stationary, $w = \pi \otimes \pi$, $w' = \pi' \otimes \pi'$, $r = w'/w$. For the pair chain,

$$\sum_s (w'Q)(s) \log r(s) - \sum_s r(s)(wQ)(s) = -\sum_{s\neq s'} w(s')Q(s',s)\Big[r(s) - r(s') - r(s') \log \tfrac{r(s)}{r(s')}\Big] \leq 0,$$

with equality only if r is constant. By (i) and since $\log r(a,b) = \ell(a) + \ell(b)$, $\ell = \log(\pi'/\pi)$, with a, b i.i.d. under w', the left-hand side vanishes by symmetry. So $\pi' = \pi$. □

On the tree the weights are $(-2, +1)$ and the same computation gives $-\mathbb{E}_{\pi'}[v'\ell] + \mathbb{E}_{\pi'}[v]$, which need not vanish. A linear-programming experiment (`lp_tree_vs_line.py` in the `code` folder) finds positive rates making two random product laws balanced in 199 of 200 cases on the tree and in none on the line.

Mechanism. Branching lets the defect $v(b) - 2v(a)$ sum to zero over finite sets without being a coboundary; on $\mathbb{Z}^d$ boundaries are negligible and relative-entropy arguments (Holley; Jahnel–Külske) pin down translation-invariant stationary measures, but on the non-amenable tree they break down. Two stationary product measures with different densities then coexist, and their mixture remembers in its tail which one is present. Positive rates give local irreducibility ("ellipticity"), but quasilocality is about the influence of infinity, which is governed by the geometry.

Which hypothesis breaks; what is not shown. The graph is a tree; the rates use an orientation towards a fixed end; both endpoints of an edge may change in one jump (single-site updates are not covered); the non-quasilocal measure is a mixture of two product measures. The question on $\mathbb{Z}^d$ remains open. Follow-up questions: see `followups.md` in the entry folder.

References

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